

Lecture 6: Excessive Volatility in Prices and Returns

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Intro

**Do Stock Prices Move Too Much to be Justified by
Subsequent Changes in Dividends?**

Motivation

- Shiller ([1981](#)) is a provocative paper
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- D_{t+k} = dividend at time $t+k$
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- D_{t+k} = dividend at time $t+k$
- r = constant discount rate
- With constant r , can this model generate the same volatility in prices as seen in the data?
- Price changes **must** be driven by news about future dividends!

- The “efficient market model” implies:

$$P_t = \mathbb{E}_t \left[\sum_{k=1}^{\infty} \frac{D_{t+k}}{(1+r)^k} \right] = \sum_{k=1}^{\infty} \gamma^{k+1} \mathbb{E}_t[D_{t+k}], \quad \gamma \equiv \frac{1}{1+r}$$

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- Realized dividends are the sum of expected dividends and an “innovation” (forecast error):

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- The realized series cannot be smoother than its projection:

$$\sigma^2(D_{t+k}) = \sigma^2(\mathbb{E}_t[D_{t+k}] + \varepsilon_{t+k}) = \sigma^2(\mathbb{E}_t[D_{t+k}]) + \sigma^2(\varepsilon_{t+k}) \geq \sigma^2(\mathbb{E}_t[D_{t+k}])$$

Shiller's Key Insight

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- Let Π_t be the GDP deflator and $A \equiv 1 + g$, where g a growth rate for real dividends
- Shiller deflated the index prices and dividends by inflation and the growth rate:

$$p_t \equiv \frac{P_t}{\Pi_t A^{t-T}}, \quad d_t \equiv \frac{D_t}{\Pi_t A^{t+1-T}}$$

where $T = 1978$ is the last year of the sample and g is estimated as the constant log-growth rate of real dividends.

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- Then we have $p_t = \sum_{k=1}^{\infty} \tilde{\gamma}^{k+1} \mathbb{E}_t[d_{t+k}]$, where $\tilde{\gamma} \equiv \frac{A}{1+r} = A\gamma$

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- Then we have $p_t = \sum_{k=1}^{\infty} \tilde{\gamma}^{k+1} \mathbb{E}_t[d_{t+k}]$, where $\tilde{\gamma} \equiv \frac{A}{1+r} = A\gamma$
- Similarly: $p_t^* = \sum_{k=1}^{\infty} \tilde{\gamma}^{k+1} d_{t+k}$

Shiller's Key Insight

- Crucially, we have the relationship:

$$p_t = \mathbb{E}_t[p_t^*] \implies \sigma^2(p_t) = \sigma^2(\mathbb{E}_t[p_t^*]) \leq \sigma^2(p_t^*)$$

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Implementation matters:

- Dividends after 1978 are unknown at time t
- Shiller assumed that they would grow at rate g forever
- This assumption likely understates $\sigma^2(p_t^*)$

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- Dividends after 1978 are unknown at time t
- Shiller assumed that they would grow at rate g forever
- This assumption likely understates $\sigma^2(p_t^*)$
- p_{1978}^* is computed with “fake dividends” and then you use the recursion backward:

$$p_t^* = \tilde{\gamma}(d_{t+1} + p_{t+1}^*)$$

- The discount rate r is such that $r = \frac{\mathbb{E}[d]}{\mathbb{E}[p]}$ (why was deflating necessary for this step?)

The Main Plot

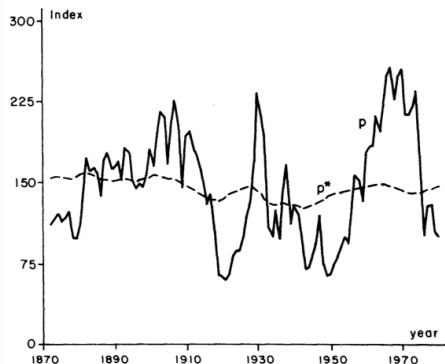


FIGURE 1

Note: Real Standard and Poor's Composite Stock Price Index (solid line p) and *ex post* rational price (dotted line p^*), 1871–1979, both detrended by dividing a long-run exponential growth factor. The variable p^* is the present value of actual subsequent real detrended dividends, subject to an assumption about the present value in 1979 of dividends thereafter. Data are from Data Set 1, Appendix.

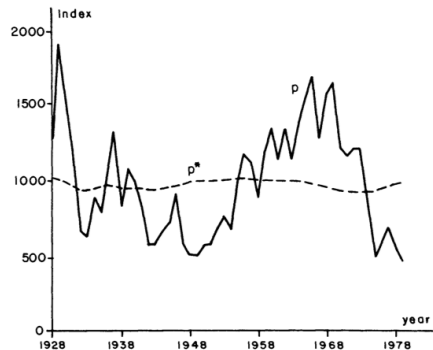


FIGURE 2

Note: Real modified Dow Jones Industrial Average (solid line p) and *ex post* rational price (dotted line p^*), 1928–1979, both detrended by dividing by a long-run exponential growth factor. The variable p^* is the present value of actual subsequent real detrended dividends, subject to an assumption about the present value in 1979 of dividends thereafter. Data are from Data Set 2, Appendix.

Main Numerical Result

- Strange things: inequalities upside down!
- You see many rows because he derived many bounds with different assumptions
- The data is rejecting theory by an order of magnitude
- $50 > 8$ and $355.9 > 26.8$

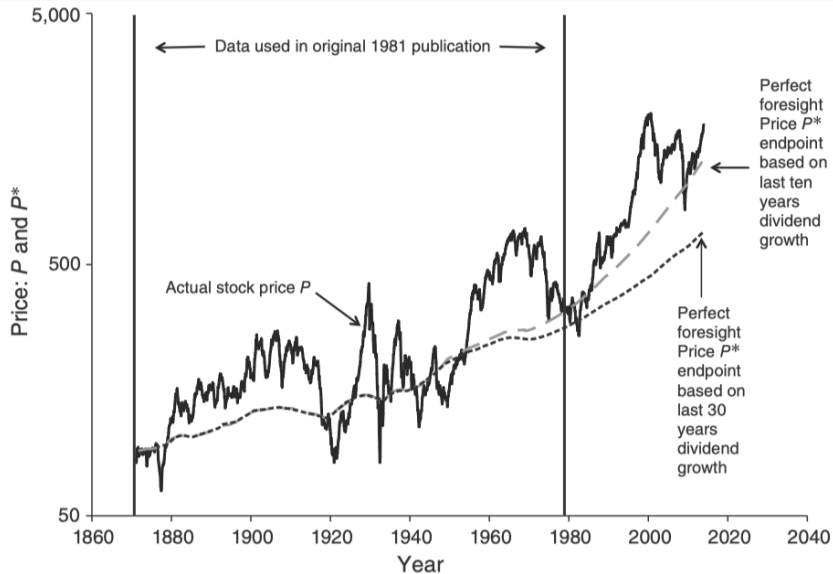
Historical evidence:

- During the Great Depression, (real) dividends were below average only on 1933-1935, and 1939
- But returns were pretty bad!

TABLE 2—SAMPLE STATISTICS FOR PRICE AND DIVIDEND SERIES

	Data Set 1: Standard and Poor's	Data Set 2: Modified Dow Industrial
Sample Period:	1871–1979	1928–1979
1) $E(p)$	145.5	982.6
$E(d)$	6.989	44.76
2) $\bar{r}$	.0480	0.456
$\bar{r}_2$	.0984	.0932
3) $b = \ln \lambda$	.0148	.0188
$\hat{\sigma}(b)$	(.0011)	(1.0035)
4) $cor(p, p^*)$	.3918	.1626
$\sigma(d)$	1.481	9.828
Elements of Inequalities:		
Inequality (I)		
5) $\sigma(p)$	50.12	355.9
6) $\sigma(p^*)$	8.968	26.80
Inequality (II)		
7) $\sigma(\Delta p + d_{-1} - \bar{r}p_{-1})$	25.57	242.1
$min(\sigma)$	23.01	209.0
8) $\sigma(d)/\sqrt{\bar{r}_2}$	4.721	32.20
Inequality (13)		
9) $\sigma(\Delta p)$	25.24	239.5
$min(\sigma)$	22.71	206.4
10) $\sigma(d)/\sqrt{2\bar{r}}$	4.777	32.56

Newer Data



What if Discount Rates Vary?

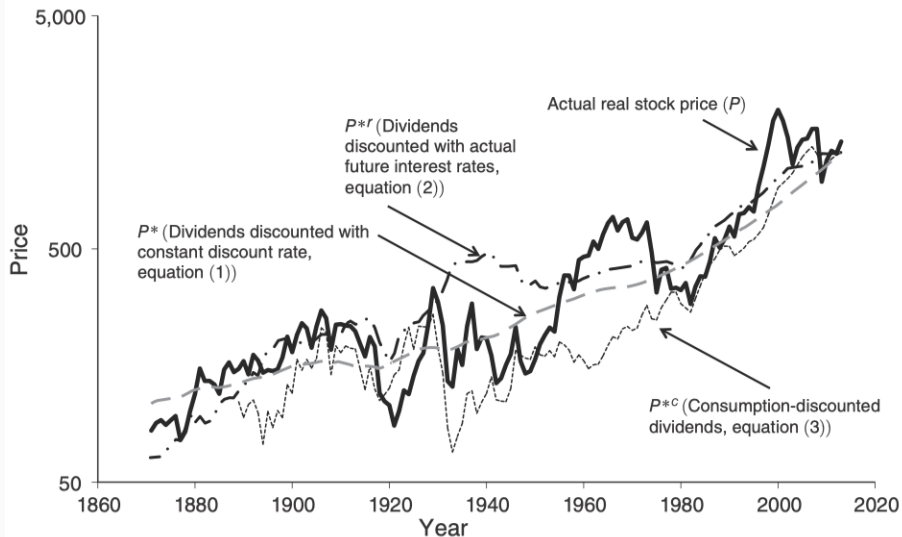
- We can always write:

$$P_t = \mathbb{E}_t \left[\sum_{k=1}^{\infty} D_{t+k} \prod_{j=1}^k \frac{1}{1 + r_{t+j-1}} \right]$$

- If we impose no discipline, this will always be true for some series of returns
- Shiller ([1981](#)) used 6-month prime commercial paper + a constant
- Shiller ([2014](#)) used our good and old friend:

$$r_{t+1} = -\log(\beta) + \delta \cdot \Delta c_{t+1}$$

Newer Data and Time-Varying Interest Rates



Where The Literature Took This

Takeaway message:

- Stock prices move too much to be justified by subsequent changes in dividends
- The natural way to reconcile this is allowing for rich time-variation in discount rates

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1. **Return decompositions:** how much of price movements is due to changes in dividend expectations vs expected returns? ([Campbell and Shiller 1988](#); [Campbell 1991](#); [Cochrane 2011](#))

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Two Paths Forward:

1. **Return decompositions:** how much of price movements is due to changes in dividend expectations vs expected returns? ([Campbell and Shiller 1988](#); [Campbell 1991](#); [Cochrane 2011](#))
2. **Behavioral finance and limits to arbitrage:** prices deviate from fundamentals due to investor psychology and limits to arbitrage ([De Long et al. 1990](#); [Shleifer and Vishny 1997](#); [Shiller 2014](#); [Barberis and Thaler 2003](#); [Barberis 2018](#))

- Excess volatility could reflect investor psychology, not rational risk premia
- Noise traders and limits to arbitrage ([De Long et al. 1990](#); [Shleifer and Vishny 1997](#))
- Shiller ([2014](#)): irrational exuberance and social dynamics drive speculative prices

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- Shiller ([2014](#)): irrational exuberance and social dynamics drive speculative prices
- Surveys: Barberis and Thaler ([2003](#)) and Barberis ([2018](#)) (extrapolation, overconfidence, prospect theory)
- We will not pursue this path, but it is an important and active research program

Questions?

Discount Rates: They Like to Move It!

The Campbell-Shiller Decomposition

- Campbell and Shiller (1988) made a simple, crazy powerful point
- Prices are *discounted expected cash-flows*: either return or dividends expectations must move to **generate** returns!

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- Prices are *discounted expected cash-flows*: either return or dividends expectations must move to **generate** returns!
- *The dividend-price ratio can only move if either **expected returns** or **expected dividend growth** moves, regardless of your mambo-jambo model!*
- They introduced the “Campbell-Shiller” decomposition, based on a Taylor expansion
- An almost model-free playground to test theories of time-varying discount rates

Campbell–Shiller Log-Linearization (Setup)

- Exact log return identity:

$$h_t \equiv \log(P_{t+1} + D_t) - \log(P_t)$$

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- Rewrite the nonlinear term:

$$\log(P_{t+1} + D_t) = p_{t+1} + \log(1 + \exp(d_t - p_{t+1}))$$

- Define the log dividend–price ratio:

$$\delta_{t+1} \equiv d_t - p_{t+1}$$

- Dividend D_t is paid at time $t + 1$ in this notation (keeping it close to the paper)

Campbell–Shiller Log-Linearization (Taylor Step)

- Let $g(\delta) \equiv \log(1 + e^\delta)$ and expand around $\bar{\delta}$:

$$g(\delta_{t+1}) \approx g(\bar{\delta}) + g'(\bar{\delta})(\delta_{t+1} - \bar{\delta})$$

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- Derivative and the key constant ρ :

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- Collect constants into k :

$$\log(P_{t+1} + D_t) \approx k + \rho p_{t+1} + (1 - \rho) d_t$$

Campbell–Shiller Log-Linear Return + Forward Solution

- Substitute into the return identity:

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- Express in terms of $\delta_t \equiv d_{t-1} - p_t$:

$$h_t \approx k + \Delta d_t + \delta_t - \rho \delta_{t+1}$$

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- Rearranged recursion and forward solution:

$$\delta_t = (h_t - \Delta d_t - k) + \rho \delta_{t+1} \Rightarrow \boxed{\delta_t = \sum_{j=0}^{\infty} \rho^j (h_{t+j} - \Delta d_{t+j}) - \frac{k}{1 - \rho}}$$

- This equation has no economic content: it is an (approximate) **identity**. Intuition?

How to Add Empirical Content

- Notice that we can take expectations conditional on information at time t :

$$\delta_t = \mathbb{E}_t \left[\sum_{j=0}^{\infty} \rho^j (h_{t+j} - \Delta d_{t+j}) \right] - \frac{k}{1-\rho} = \sum_{j=0}^{\infty} \rho^j (\mathbb{E}_t[h_{t+j}] - \mathbb{E}_t[\Delta d_{t+j}]) - \frac{k}{1-\rho}$$

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- Example: suppose that for some r_{t+j} and some c we have $\mathbb{E}_t[h_{t+j}] = \mathbb{E}_t[r_{t+j}] + c$
- We can write:

$$\delta_t = \sum_{j=0}^{\infty} \rho^j (\mathbb{E}_t[r_{t+j}] - \mathbb{E}_t[\Delta d_{t+j}]) + \frac{c-k}{1-\rho}$$

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- This will become testable once we specify a model for returns and dividends
- Example: give me $\tilde{r}_{t,j} \equiv \mathbb{E}_t[r_{t+j}]$ and $\tilde{\Delta d}_{t,j} \equiv \mathbb{E}_t[\Delta d_{t+j}]$ and generate $\tilde{\delta}_t$
- We should have $\delta_t \approx \tilde{\delta}_t$

Questions?

Campbell and Shiller Meet the Data

There is hope in the approximation!

Table 2
Summary statistics for stock market data

Statistic	Data set and sample period			Correlation of Cowles/S&P and NYSE, 1926–1986
	Cowles/S&P, 1871–1986	Cowles/S&P, 1926–1986	NYSE, 1926–1986	
Δp_t				
Mean	0.032	0.044	0.042	0.972
Standard deviation	0.178	0.200	0.208	
Δd_t				
Mean	0.030	0.041	0.040	0.958
Standard deviation	0.132	0.131	0.134	
δ_t				
Mean	−3.053	−3.121	−3.143	0.985
Standard deviation	0.277	0.294	0.290	

All variables in this table are nominal and measured annually. p_t is the log stock price, d_t is the log dividend, and δ_t is the log dividend-price ratio $d_{t-1} - p_t$.

Setup

- Assume that market participants use state variables y_t to do forecasts
- y_t evolves linearly (think about an $\text{MA}(\infty)$ process)
- The econometrician observes (with abuse of notation) $x_t \equiv (\delta_t, h_{t-1}, \Delta d_{t-1})^\top \subset y_t$

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- Assume that x_t (already demeaned) follows a $\text{VAR}(p)$:

$$x_t = C_1 x_{t-1} + \dots + C_p x_{t-p} + u_t$$

where all variables are demeaned and u_t is a white noise process each C_i is a 3×3 matrix

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- Rewrite the VAR in companion form:

$$z_t \equiv \begin{bmatrix} x_t \\ x_{t-1} \\ \vdots \\ x_{t-p+1} \end{bmatrix} = \begin{bmatrix} C_1 & C_2 & \dots & C_p \\ I & 0 & \dots & 0 \\ 0 & I & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ x_{t-2} \\ \vdots \\ x_{t-p} \end{bmatrix} + \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix} = A z_{t-1} + v_t$$

The VAR Approach

- The companion form implies: $\mathbb{E}_t[z_{t+j}] = A^j z_t$
- The VAR is a forecasting machine to generate discount rate and dividend forecasts
- Given A , we can compute:

$$\tilde{\delta}_t = \sum_{j=0}^{\infty} \rho^j (\mathbb{E}[h_{t+j} | \{x_t, x_{t-1}, \dots\}] - \mathbb{E}[\Delta d_{t+j} | \{x_t, x_{t-1}, \dots\}])$$

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- Let $e_1 = (1, 0, 0, \dots, 0)^\top$ and $e_2 = (0, 1, 0, \dots, 0)^\top$ and $e_3 = (0, 0, 1, 0, \dots, 0)^\top$. Then:

$$\tilde{\delta}_t = \sum_{j=0}^{\infty} \rho^j (e_2^\top A^{j+1} z_t - e_3^\top A^{j+1} z_t) = \sum_{j=0}^{\infty} \rho^j ((e_2 - e_3)^\top A^{j+1} z_t)$$

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- What condition would make the observed δ_t and the synthetic $\tilde{\delta}_t$ match?

Rational Pricing Implies Objective Forecasts

- If we impose $\delta_t = \tilde{\delta}_t$, then:

$$e_1^\top z_t = \delta_t = \tilde{\delta}_t = \sum_{j=0}^{\infty} \rho^j ((e_2 - e_3)^\top A^{j+1} z_t) = \left(\sum_{j=0}^{\infty} \rho^j (e_2 - e_3)^\top A^{j+1} \right) z_t, \forall z_t$$

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- This would imply the following restriction on the VAR coefficients:

$$e_1^\top = \left(\sum_{j=0}^{\infty} \rho^j (e_2 - e_3)^\top A^{j+1} \right) = (e_2 - e_3)^\top A (I - \rho A)^{-1}$$

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$$e_1^\top z_t = \delta_t = \tilde{\delta}_t = \sum_{j=0}^{\infty} \rho^j ((e_2 - e_3)^\top A^{j+1} z_t) = \left(\sum_{j=0}^{\infty} \rho^j (e_2 - e_3)^\top A^{j+1} \right) z_t, \forall z_t$$

- This would imply the following restriction on the VAR coefficients:

$$e_1^\top = \left(\sum_{j=0}^{\infty} \rho^j (e_2 - e_3)^\top A^{j+1} \right) = (e_2 - e_3)^\top A (I - \rho A)^{-1}$$

- We can do a Wald test! This is just a (non-linear) restriction on the VAR coefficients!
- A rejection would imply that whatever theory we imposed to generate the forecasts is not consistent with the data
- $\rho = 0.93$ to match average dividend–price ratio in the data

Tested Theory 1

- They assume a constant conditional expected return $\mathbb{E}_t[h_{t+j}] = \bar{r}$
- They can use $x_t = (\delta_t, \Delta d_{t-1})$ here since $\bar{r}$ gets absorbed

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VAR estimation:

δ_{t+1}	$\begin{bmatrix} 0.706 \\ (0.066) \end{bmatrix}$	$\begin{bmatrix} 0.259 \\ (0.139) \end{bmatrix}$	0.515	0.000
Δd_t	$\begin{bmatrix} -0.197 \\ (0.039) \end{bmatrix}$	$\begin{bmatrix} 0.231 \\ (0.083) \end{bmatrix}$	0.227	0.000

Implications of VAR estimates:

$$\delta'_t = 0.636\delta_t - 0.097\Delta d_{t-1}$$

(0.123) (0.106)

$$\sigma(\delta'_t)/\sigma(\delta_t) = 0.637 \quad \text{corr}(\delta'_t, \delta_t) = 0.997$$

(0.124) (0.006)

Significance level for Wald test that $\delta'_t = \delta_t$: 0.005

- The idea of constant expected returns is easily rejected!

Tested Theory 2

- They impose a constant *premium* over the risk-free rate: $\mathbb{E}_t[h_{t+j}] = \mathbb{E}_t[r_{t+j}^f] + c$
- Use $x_t = (\delta_t, r_{t-1}^f - \delta d_{t-1})$ with nominal rvariables – inflation cancels out!

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Table 5
Testing constant expected real and excess returns

	Model version 1 (constant expected real returns)			Model version 2 (constant expected excess returns)		
	Lag length			Lag length		
	1	3	5	1	3	5
<u>Cowles/S&P, 1871–1986</u>						
h_t regression test	0.078	0.056	0.083	0.100	0.007	0.014
ξ_t regression test	0.045	0.035	0.055	0.073	0.005	0.009
Test that $\delta'_t = \delta_t$	0.005	0.000	0.000	0.009	0.000	0.000
$\sigma(\delta'_t)/\sigma(\delta_t)$	0.637	0.370	0.382	0.674	0.434	0.402
	(0.124)	(0.129)	(0.073)	(0.111)	(0.143)	(0.060)
corr (δ'_t, δ_t)	0.997	0.837	0.326	0.999	0.856	0.431
	(0.006)	(0.246)	(0.565)	(0.004)	(0.200)	(0.583)

- The third one: $\mathbb{E}_t h_{t+j} = \alpha \cdot \mathbb{E}_t [\delta c_{t+j}] + \text{constant}$, where α is an estimated RRA parameter
- The fourth one: $\mathbb{E}_t h_{t+j} = \alpha \cdot V_t + \text{constant terms}$, where $V_t = h_t^2$ is a proxy for volatility
- This is due to Mehra and Prescott (1985) and Pindyck (1984)

Tested Theory 3 and 4

Table 7
Testing consumption- and volatility-based models of the discount rate

	Model version 3 (consumption)		Model version 4 (volatility)	
	Lag length		Lag length	
	1	3	1	3
<u>Cowles/S&P, 1871–1986</u>				
Estimate of α	-2.191	-0.423	2.552	0.960
(standard error)	(1.399)	(0.818)	(2.549)	(0.985)
ξ_t regression test	0.072	0.038	0.170	0.096
Test that $\delta'_t = \delta_t$	0.012	0.000	0.039	0.007
$\sigma(\delta_t - \delta'_{dt})/\sigma(\delta_t)$	0.484	0.711	0.368	0.724
	(0.100)	(0.216)	(0.122)	(0.211)
$\sigma(\delta'_{rt})/\sigma(\delta_t)$	0.227	0.038	0.077	0.095
	(0.149)	(0.081)	(0.116)	(0.148)
corr ($\delta_t - \delta'_{dt}$, δ'_{rt})	0.910	-0.024	0.339	0.660
	(0.073)	(0.788)	(1.415)	(0.525)

Main Takeaways

- Campbell and Shiller (1988) called it: the dividend-price ratio can only move if either expected returns or expected dividend growth moves
- Their decomposition imposes very little structure! *You give me some theory, my VAR gives you a test.*
- Constant expected returns cannot rationalize the data!
- Allowing the discount rates to depend on consumption and volatility was *not* enough

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